Program: **Diploma in Mechatronics**	
Course Code: **5438**	Course Title: **Computer Integrated Design and Manufacturing lab**
Semester: **5**	Credits: **2**
Course Category: **Program Elective**	
Periods per week: **3 (L:0 T:1 P:2)**	Periods per semester: **45**

Course Objectives:

- To introduce the various features in the menu of solid modeling package.
- To explain the synthesis of various parts or components in an assembly.
- To familiarize the students with CNC programs for various jobs.
- To familiarize the students with basics of 3D printing technology

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Knowledge in Manufacturing		Manufacturing technology	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the working of CNC turning and milling machine.	12	Applying
CO2	Prepare the part program using simulation software for Lathe and Milling.	9	Applying
CO3	Prepare the part program, edit and execute in CNC turning and machining Centre	12	Applying
CO4	Introduction to 3D Printing	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			3
CO3	3			3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Demonstrate the working of CNC turning and milling machine**		
M1.01	Demonstrate CNC Programming and Machining: Introduction: Study of CNC lathe, milling	2	Applying
M1.02	Interpret international standard codes: G-Codes and M-Codes	2	Applying
M1.03	Use Format – Dimensioning methods	2	Applying
M1.04	Prepare Program – Turning simulator – Milling simulator, IS practice – commands menus	2	Applying
M1.05	Modify the program in the CNC machines	2	Applying
M1.06	Apply the program in the CNC machines; Exercises: Note: Print the Program from the Simulation Software and make the Component in the CNC Machine.	2	Applying
CO2	**Prepare the part program using simulation software for Lathe and Milling.**		
M2.01	Produce component in CNC Turning Machine with part program (Linear and Circular interpolation, Material: Aluminum/Acrylic/Plastic rod)	3	Applying
M2.02	Produce component with multiple turning operations in CNC Turning Machine with Part program (Stock removal cycle)	3	Applying

M2.03	Produce component with thread and grooving in the CNC Turning Machine with Part program (canned Cycle)	3	Applying
	Lab Exam – I	1.5	
CO3	**Prepare the part program, edit and execute in CNC turning and machining Centre.**		
M3.01	Produce component with grooving in CNC Milling Machine with Part program (Linear interpolation and Circular interpolation, Material: Aluminum/ Acrylic/ Plastic	6	Applying
M3.02	Produce component with drilling, tapping and countersinking in the CNC Milling Machine. (Canned cycle)	3	Applying
M3.03	Produce component with mirroring in the CNC Milling Machine with Part program (sub program)	3	Applying
CO4	**Introduction to 3D Printing**		
M4.01	Overview of 3D printing significance, introduction to technologies (FDM, SLA, SLS), and applications in industries (toys, prototypes, healthcare)	1	Understanding
M4.02	Introduction to design concepts for 3D printing, use of basic (open source) CAD software (e.g., Tinkercad, freecad) for simple 3D models, and understanding design considerations (dimensions, tolerances, aesthetics)	2	Applying
M4.03	Overview of preparation/slicing (open source) software (e.g., Cura, PrusaSlicer), basic printing settings (layer height, infill, support structures), and a step-by-step guide for preparing a print file.	3	Applying
M4.04	Practice Printing of objects with PLA/ABS materials	3	Applying
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	CAD/CAM/CIM – P. Radhakrishnan, S. Subramaniyan& V. Raju, New Age International Pvt. Ltd., New Delhi, 3rd Edition,

R1	Manufacturing process for engineering materials – SeropeKalpakjian
R2	N.C. Machines and CAM - Pressman Williams

Online Resources:

Sl.No	Website Link
1	https://www.engineeringbookspdf.com/download/?file=4559
2	https://www.instructables.com/id/UNDERSTANDING-CNC-AND-BASIC-PART-PROGRAMMING-FOR-M/
3	https://3dprintingindustry.com/3d-printing-basics-free-beginners-guide#04-processes